

Cybersecurity Basics

1. CIA Triad

Confidentiality

Confidentiality ensures that information is accessible only to authorized individuals. It protects sensitive data from unauthorized access.

Example: Password-protected accounts and encrypted files.

Integrity

Integrity ensures that data remains accurate, complete, and unchanged unless modified by authorized users.

Example: File hashing and digital signatures.

Availability

Availability ensures that systems, applications, and data are accessible whenever needed by authorized users.

Example: Backup servers and disaster recovery systems.

2. Threat Types

Phishing

A cyberattack where attackers use fake emails, websites, or messages to steal sensitive information such as passwords and banking details.

Malware

Malicious software designed to damage, disrupt, or gain unauthorized access to computer systems.

DDoS (Distributed Denial of Service)

An attack that overwhelms a server or network with excessive traffic, making services unavailable to legitimate users.

SQL Injection

A web attack that inserts malicious SQL commands into application inputs to access or manipulate databases.

Brute Force Attack

A method of gaining access by repeatedly trying different password combinations until the correct one is found.

Ransomware

A type of malware that encrypts files or locks systems and demands payment to restore access.

3. Attack Vectors

Social Engineering

Manipulating people into revealing confidential information or performing actions that compromise security.

Wireless Attacks

Attacks targeting wireless networks, including unauthorized access to Wi-Fi networks and interception of wireless communications.

Insider Threats

Security risks originating from employees, contractors, or other trusted individuals who misuse their access privileges.

Conclusion

Cybersecurity focuses on protecting systems, networks, and data from digital attacks. Understanding the CIA Triad, common threats, and attack vectors forms the foundation of cybersecurity knowledge.